Major Problems in Recurrent Neural Networks (RNNs)

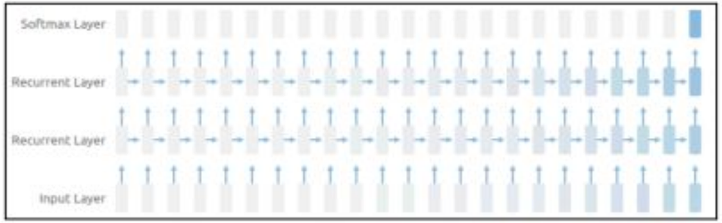
Main Problems in RNNs

- **Vanishing Gradient Problem:** Gradients become very small during backpropagation, causing slow learning and difficulty in capturing long-term dependencies.
- **Exploding Gradient Problem:** Gradients become very large, leading to unstable training and potential divergence of the model.
- **Short-term Memory:** Difficulty in maintaining information over long sequences, causing poor performance on tasks requiring long-term context retention.
- **Handling Long Sequences:** Performance degrades with increasing sequence length, making RNNs less suitable for tasks involving long sequences.
- **Gradient Clipping Requirement:** To manage exploding gradients, gradient clipping is needed, adding complexity to the training process.

Main Problems in RNNs

Vanishing and Exploding Gradients

- Vanishing gradient problem



Contribution from the earlier steps becomes insignificant in the gradient

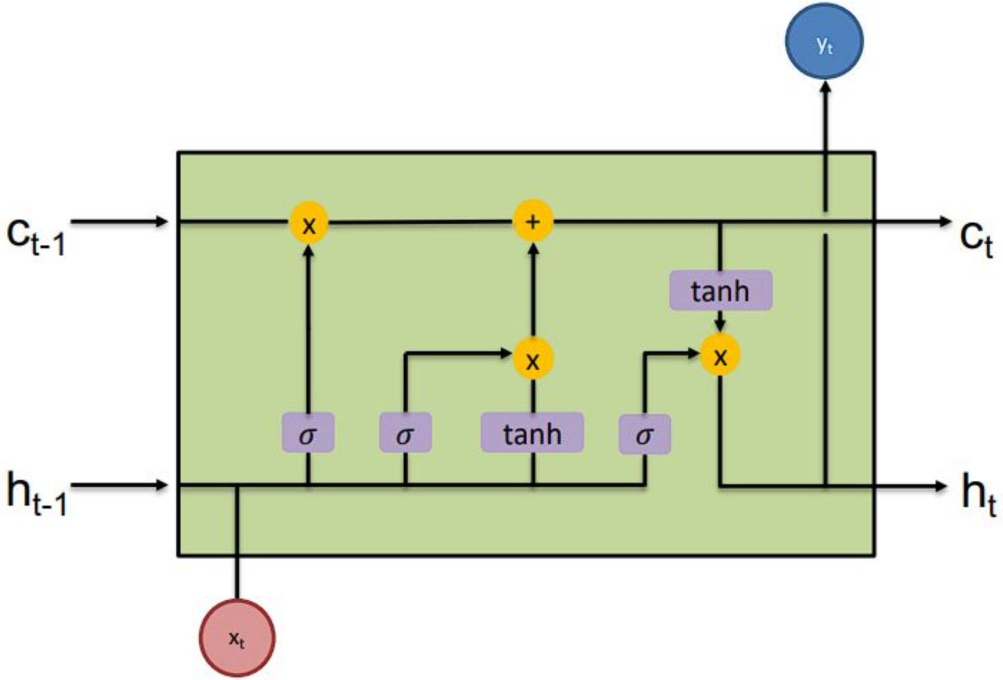
- Exploding gradient problem:
  - Gradients become very very large due to a single or multiple gradient values becoming very high.
- Vanishing gradient is more concerning
  - Exploding gradient easily solved by clipping the gradients at a predefined threshold value.
- Solution: gated RNNs
  - LSTM, GRU (Gated Recurrent Units), a variant of LSTM

Differences: RNNs Vs. LSTMs

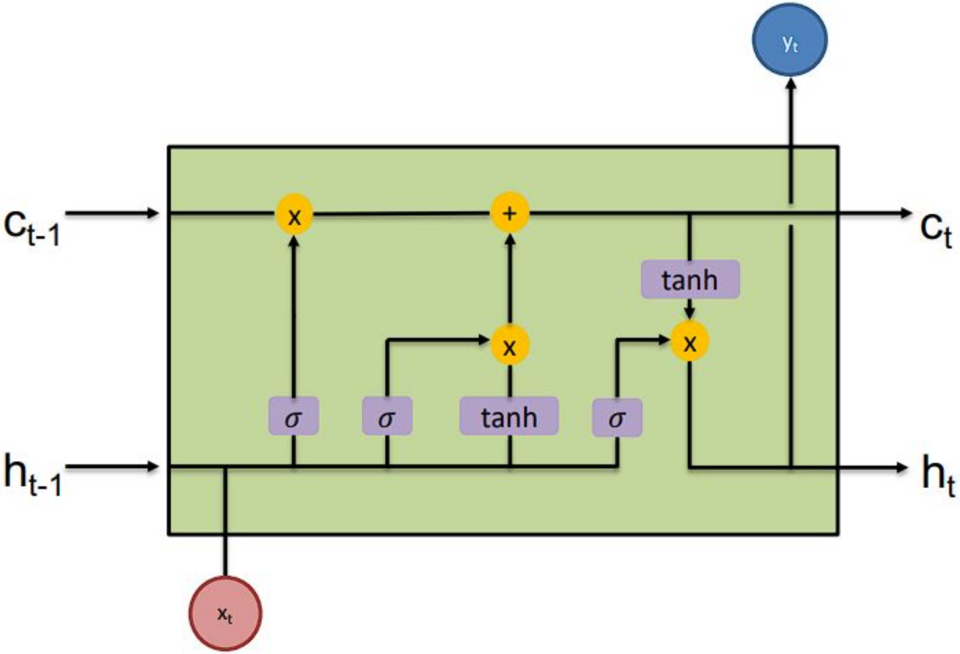
Feature	RNNs	LSTMs
Vanishing Gradient Problem	Common	Mitigated
Exploding Gradient Problem	Common	Mitigated
Memory	Short-term	Long-term
Sequence Length Handling	Poor	Better
Internal Mechanisms	Simple recurrent connections	Cell state, input, forget, output gates
Training Stability	Less stable	More stable
Task Performance	Weaker for long sequences	Stronger for long sequences

Long Short-Term Memory (LSTMs)

Long Short-Term Memory (LSTM)



LSTM: Architecture

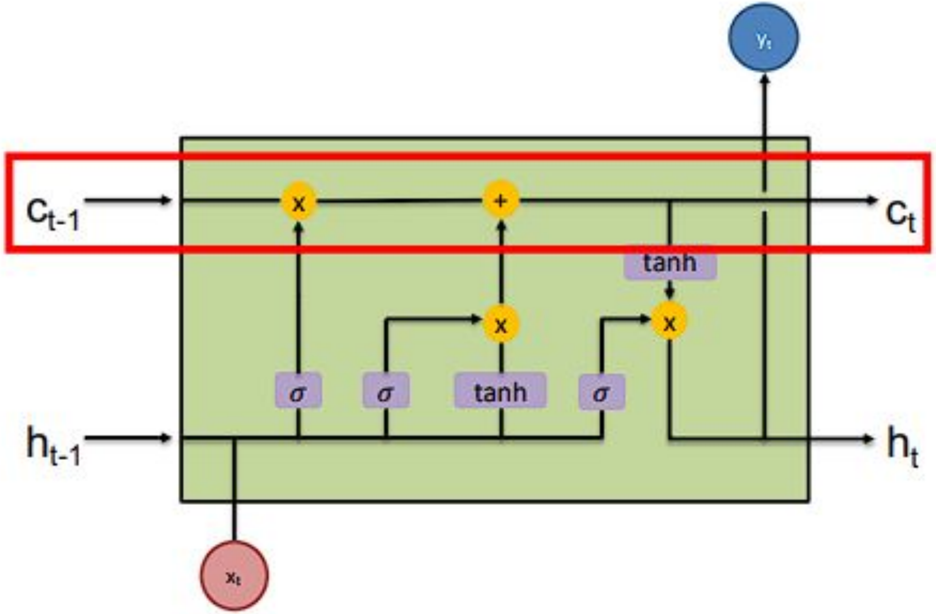


- Compared to RNNs, additionally to the **hidden state** h_t, we have another state, c_t, said the **cell state**.
- The information flow is regulated by **three gates**: the forget gate, the input gate and the output gate.

LSTM: Main Components

- The **Forget Gate** f_t decides which information to discard from the cell state.
- The **Input Gate** i_t decides which new information to add to the cell state.
- The **Candidate Cell State** $\tilde{C}_t$ represents new information that could be added to the cell state.
- The **Cell State Update** C_t combines the retained and new information to update the cell state.
- The **Output Gate** o_t decides the information that will be output from the LSTM.
- The **Hidden State Update** h_t combines the output gate decision with the updated cell state to produce the new hidden state.

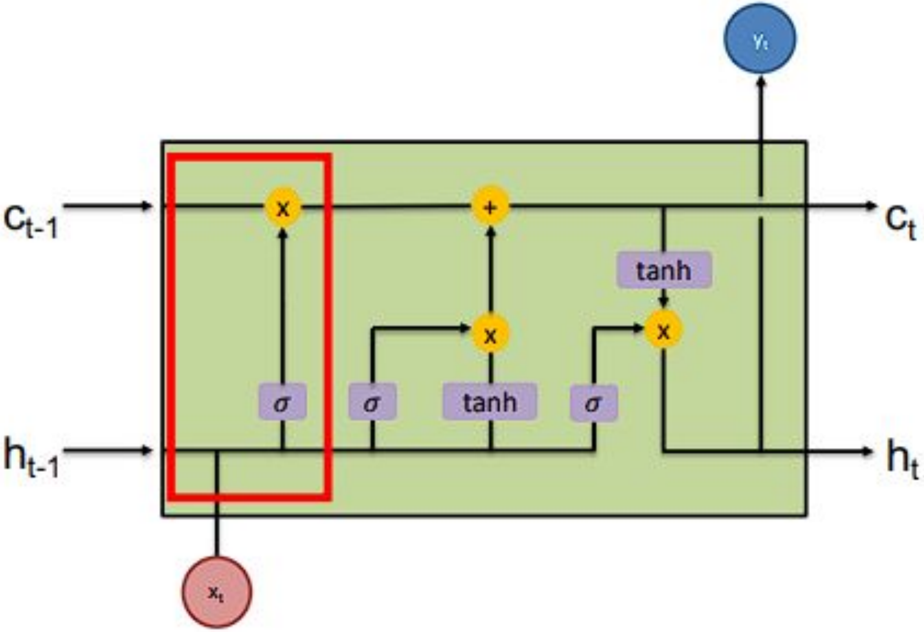
LSTM: Cell State



The cell state has:

- Only minor interactions
- Simple information flow
- Other gates regulates weather it is preserved/not-preserved or updated.

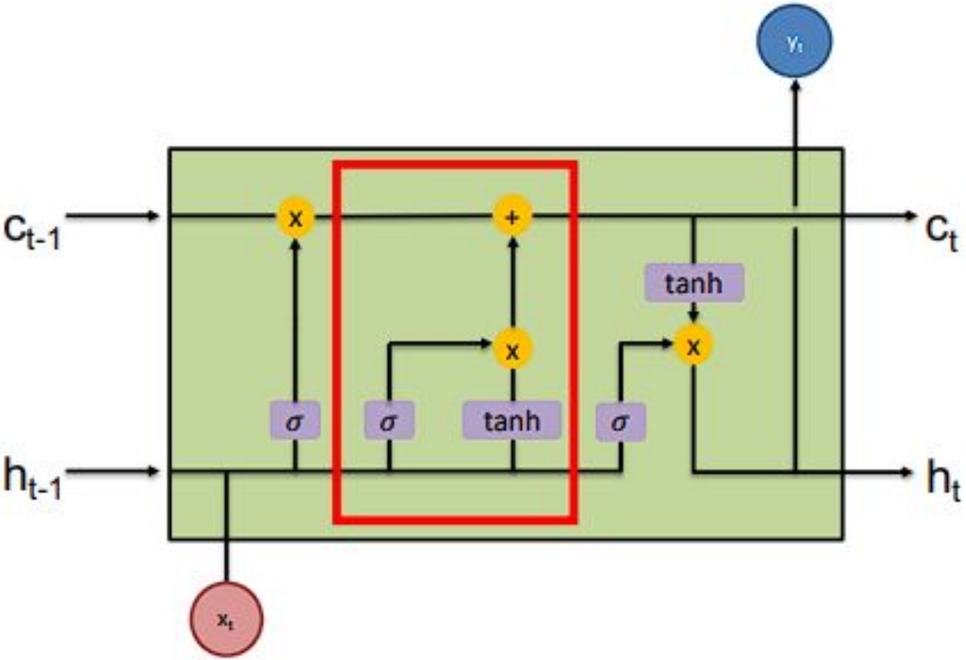
LSTM: Forget Gate



The **forget gate** decides how much information to retain from the previous cell state.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right)$$

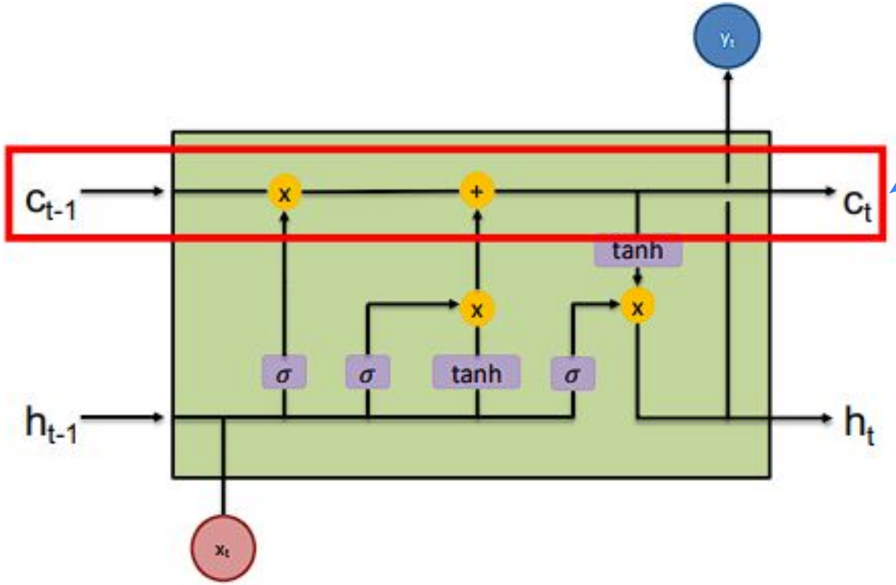
LSTM: Input Gate



The **input gate** decides the information to be added to the cell state, based on the current input.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
$$g_t = tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

LSTM: Cell State Update

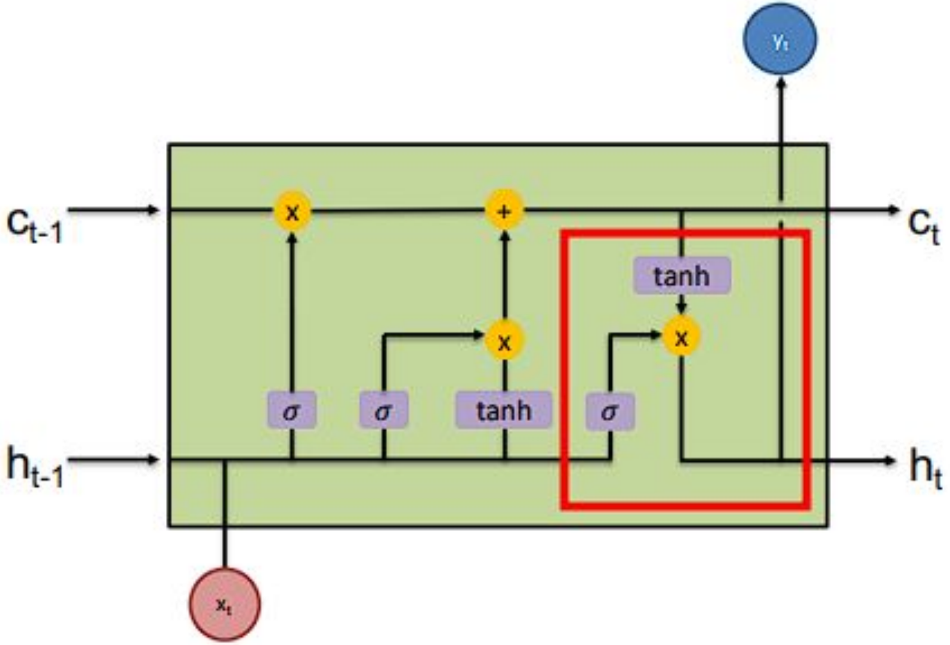


Update: Cell State

The values can be combined to update the cell state

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$

LSTM: Output Gate

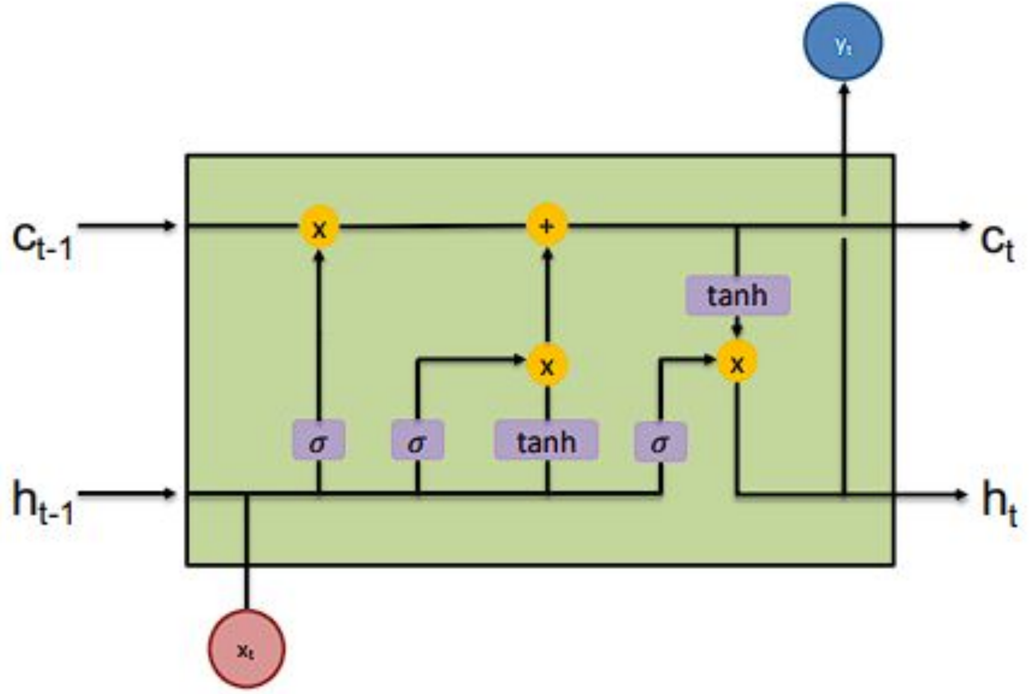


The **output gate** generates the output of the current LSTM cell, based on the current input, the previous output, and the updated cell state.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$
$$h_t = o_t * \tanh(c_t)$$

Let’s Summarize, the LSTM Cell is Described by the Following Equations!

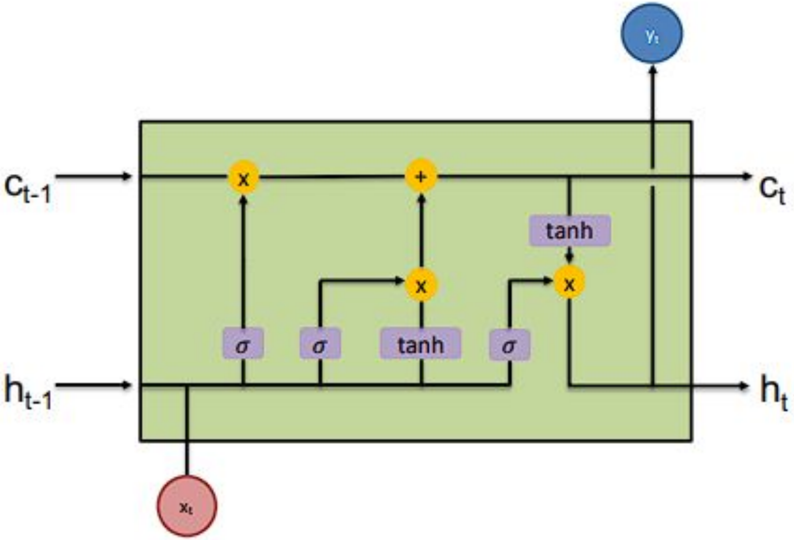
Long Short-Term Memory (LSTM)



Long Short-Term Memory (LSTM)

Component	Equation
Forget Gate (f_t)	$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$
Input Gate (i_t)	$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$
Candidate Cell State ($\tilde{C}_t$)	$\tilde{C}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$
Update Cell State (C_t)	$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$
Output Gate (o_t)	$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$
Update Hidden State (h_t)	$h_t = o_t \cdot \tanh(C_t)$
Updated Output (y_t)	$y_t = \mathrm{softmax}(W_y h_t + b_y)$

Long Short-Term Memory (LSTM)



All Equations Together:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right)$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
$$g_t = tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$
$$h_t = o_t * \tanh(c_t)$$

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Gated Recurrent Units (GRUs)

Gated Recurrent Units (GRUs)

GRUs have a simpler structure compared to LSTMs. While LSTMs have separate cell states and hidden states, GRUs combine these into a single hidden state. This reduces the complexity of the model.

Main components of Gated Recurrent Unit (GRU):

1. Reset Gate (r_t)
2. Update Gate (z_t)
3. Candidate Hidden State ($\tilde{h}_t$)

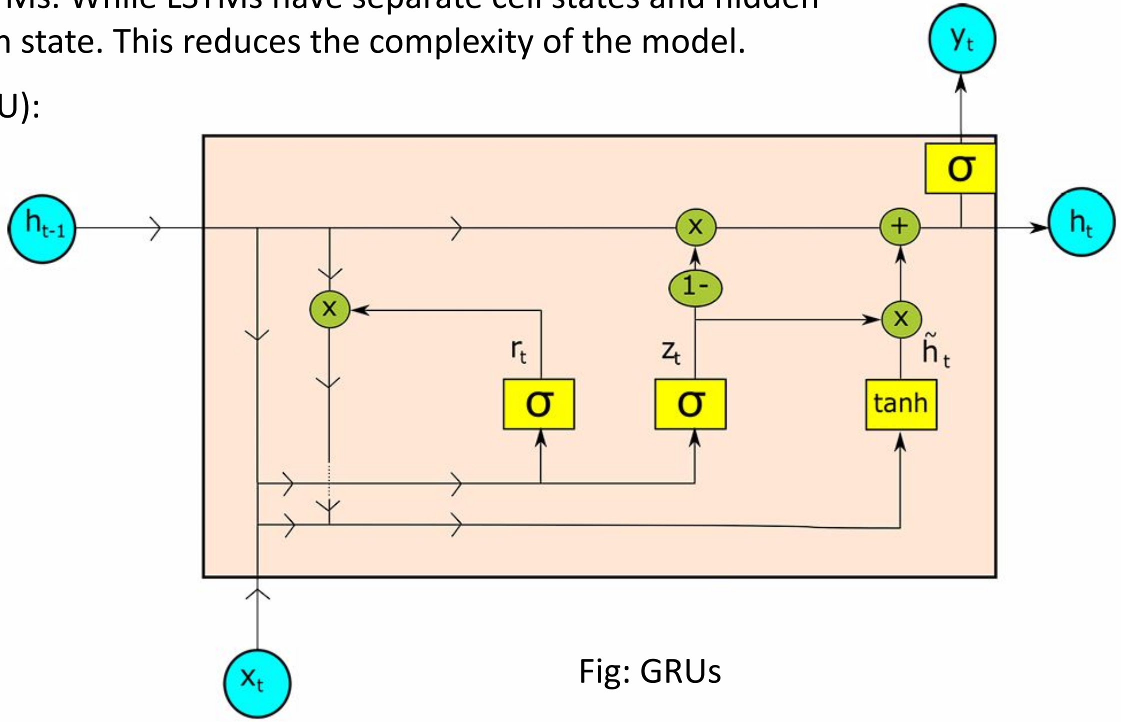

Fig: GRUs

Long Short-Term Memory (LSTM)

Main components of Long Short-Term Memory (LSTM):

1. Forget Gate (f_t)
2. Input Gate (i_t)
3. Output Gate (o_t)
4. Cell State (C_t)

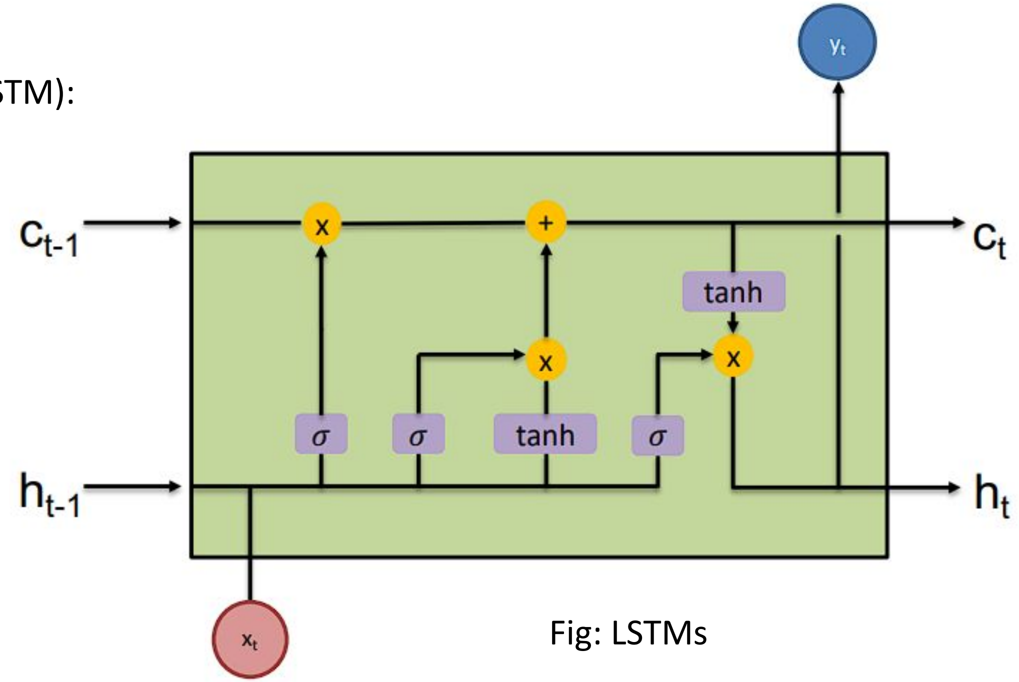

Fig: LSTMs

More Gates:

- LSTMs have **three gates**: input gate, forget gate, and output gate.
- Each gate requires its own set of weights and biases.
- This results in a **larger number of parameters** (weights and biases) that need to be stored and updated during training.

Cell State:

- LSTMs maintain a **separate cell state** in addition to the hidden state.
- The cell state requires **additional memory** to store its value for **each time step** during the forward and backward passes.

Parameters: LSTMs Vs. GRUs

- **LSTM Parameter Count:** For an LSTM cell, the parameters include weights and biases for the input, forget, and output gates, as well as the candidate cell state.
  - Total Parameters=4 × ((input_size + hidden_size) × hidden_size + hidden_size)

- **GRU Parameter Count:** For a GRU cell, the parameters include weights and biases for the reset gate, update gate, and candidate hidden state.
  - Total Parameters=3 × ((input_size + hidden_size) × hidden_size + hidden_size)

Parameters: LSTMs Vs. GRUs

Suppose we have:

- Input size $i = 100$
- Hidden state size $h = 50$

For an LSTM:

- Number of parameters: $4 \times ((100 + 50) \times 50 + 50) = 4 \times (150 \times 50 + 50) = 4 \times (7500 + 50) = 4 \times 7550 = \boxed{30200}$

For a GRU:

- Number of parameters: $3 \times ((100 + 50) \times 50 + 50) = 3 \times (150 \times 50 + 50) = 3 \times (7500 + 50) = 3 \times 7550 = \boxed{22650}$

Step by Step: Gated Recurrent Units (GRUs)

Gated Recurrent Units (GRUs)

Main components of Gated Recurrent Unit (GRU):

1. Reset Gate (r_t)
2. Update Gate (z_t)
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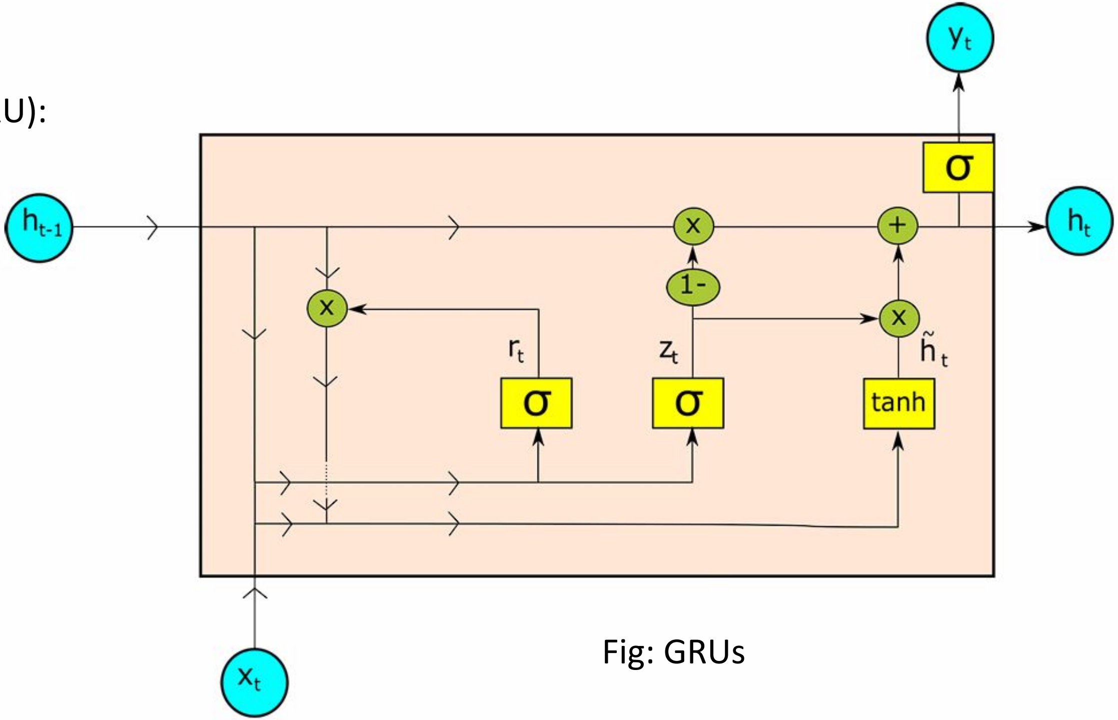


Fig: GRUs

Gated Recurrent Units (GRUs)

Steps in GRUs

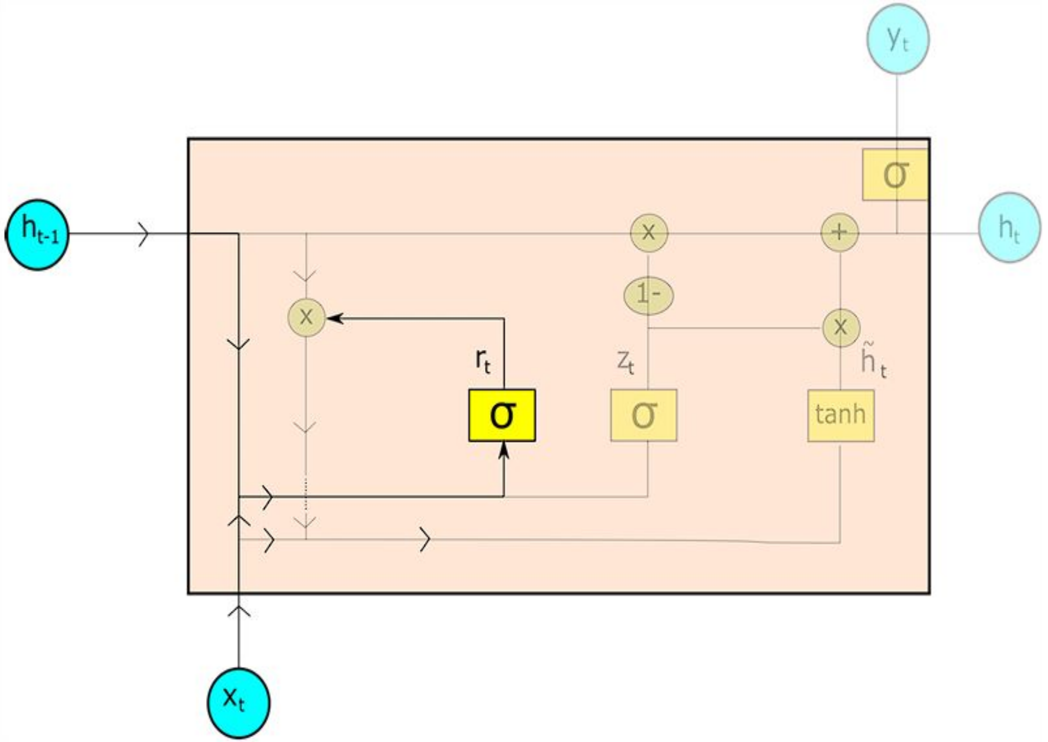
1. **Reset Gate** controls how much past information to forget.
2. **Update Gate** controls how much past information to retain.
3. **Candidate Hidden State** is computed using the reset gate's influence on the previous hidden state.
4. **Final Hidden State** is a weighted combination of the **previous hidden state** and the **candidate hidden state**, controlled by the **update gate**.

Reset Gate

Formula: $r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$

Here,

- W_r: Weight matrix for the reset gate.
- h_{t-1}: Previous hidden state.
- x_t: Current input.
- b_r: Bias term for the reset gate.

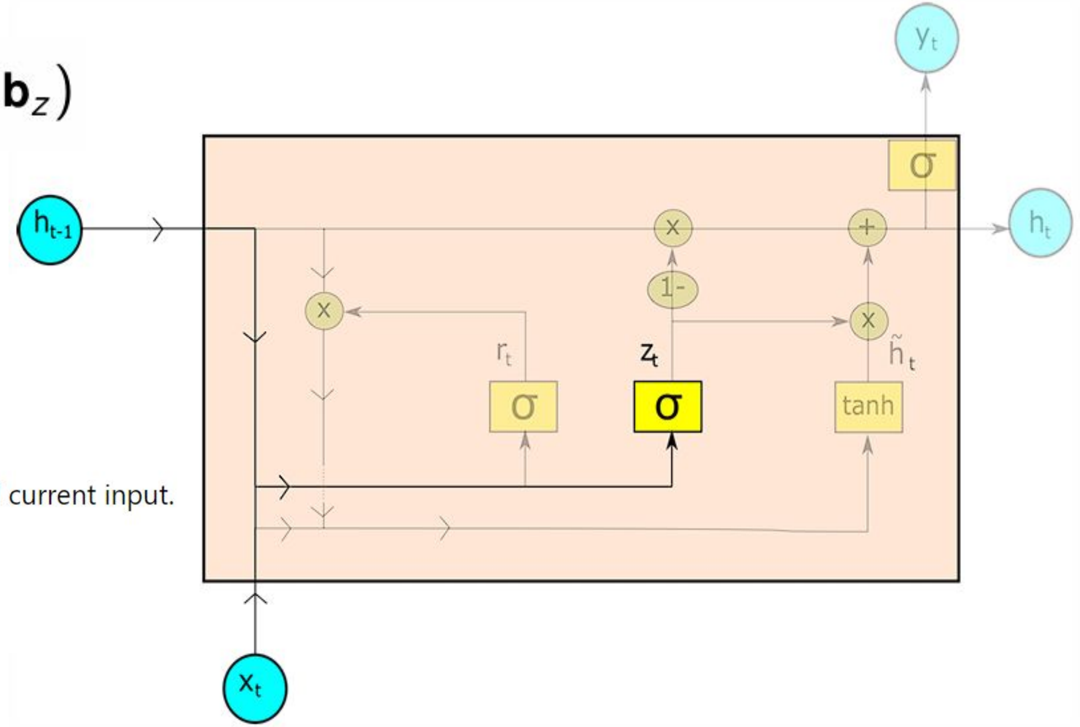


Update Gate

Formula: $\mathbf{z}_t = \sigma\left(\mathbf{W}_z \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_z\right)$

Here,

- W_z: Weight matrix for the update gate.
- h_{t-1}: Previous hidden state.
- x_t: Current input.
- h_{t-1}, x_t: Concatenation of previous hidden state and current input.
- b_z: Bias term for the update gate.
- σ: Sigmoid activation function.



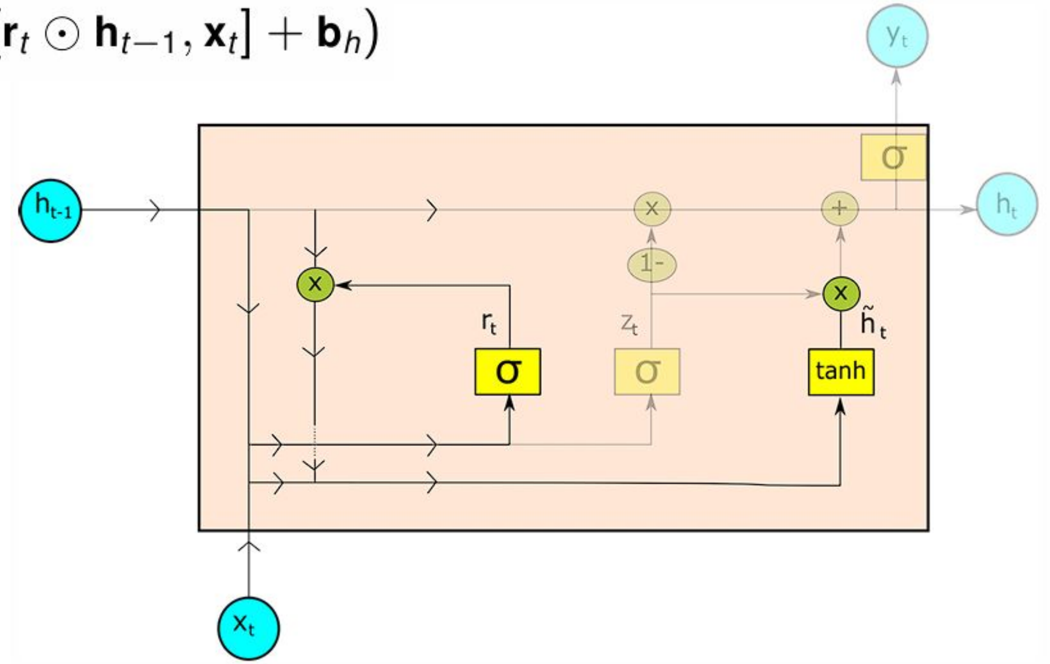
Candidate Hidden State

- Combination of input and "reset" hidden state.
- If $\mathbf{r}_t$ is close to 0 → low influence of previous hidden state

$$\tilde{\mathbf{h}}_t = tanh\left(\mathbf{W}_h \cdot [\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_h\right)$$

The Reset Gate Controls the influence of the previous hidden state

- Low r_t: Less influence of h_{t-1}.
- High r_t: More influence of h_{t-1}.

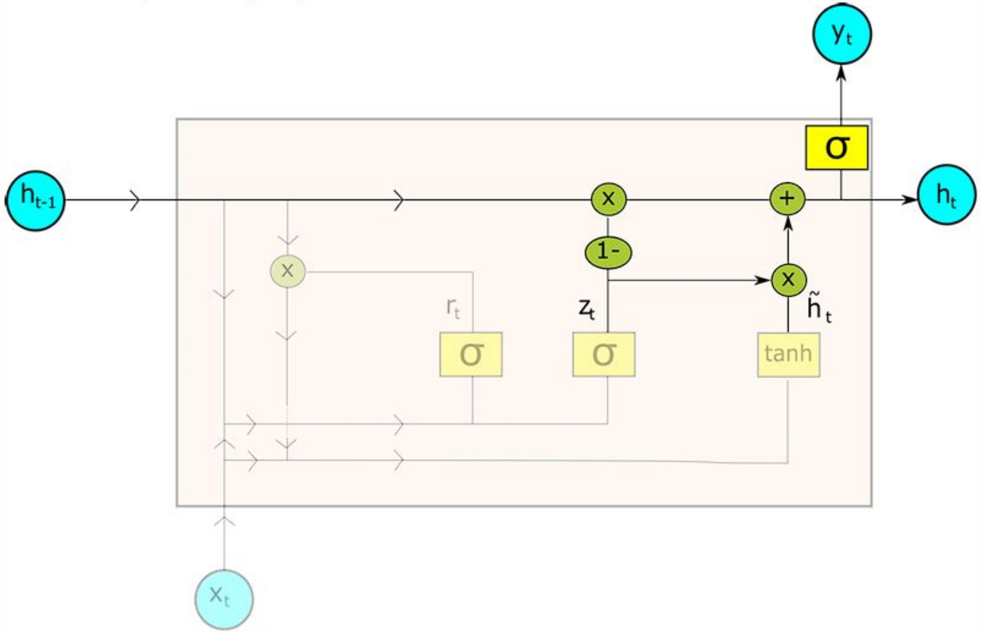


Updated Hidden State

- **Update gate** controls combination of old state and proposed update

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$

- Node output: $\hat{\mathbf{y}}_t = \sigma(\mathbf{h}_t)$



Main components of Gated Recurrent Unit (GRU):

1. Reset Gate (r_t)
2. Update Gate (z_t)
3. Candidate Hidden State ($\tilde{h}_t$)

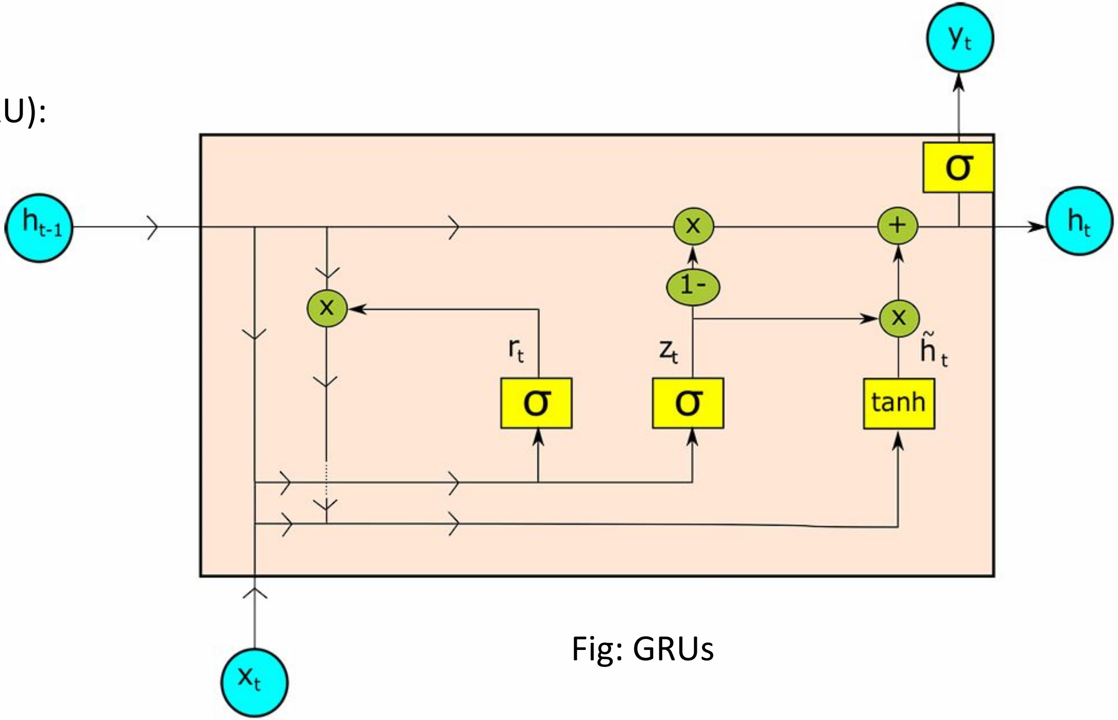


Fig: GRUs

Equations for GRU

Component	Equation
Reset Gate (r_t)	$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$
Update Gate (z_t)	$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$
Candidate Hidden State ($\tilde{h}_t$)	$\tilde{h}_t = \tanh(W_h \cdot [r_t \circ h_{t-1}, x_t] + b_h)$
Final Hidden State (h_t)	$h_t = z_t \circ h_{t-1} + (1 - z_t) \circ \tilde{h}_t$
Updated Output (y_t)	$y_t = \text{softmax}(W_y \cdot h_t + b_y)$

LSTMs Vs. GRUs

Component	LSTM	GRU
Forget Gate	$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$	Not present
Input Gate	$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$	Combined with the update gate
Output Gate	$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$	Not present
Cell State	$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$	Not present
Reset Gate	Not present	$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$
Update Gate	Not present	$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$
Candidate Cell State	$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$	$\tilde{h}_t = \tanh(W_h \cdot [r_t \circ h_{t-1}, x_t] + b_h)$
Hidden State	$h_t = o_t \cdot \tanh(C_t)$	$h_t = z_t \cdot h_{t-1} + (1 - z_t) \cdot \tilde{h}_t$